[2]

4 (a) The *I*-*V* characteristics of resistor **R** and resistor **X** are investigated. The results obtained are shown in Fig. 4.1.

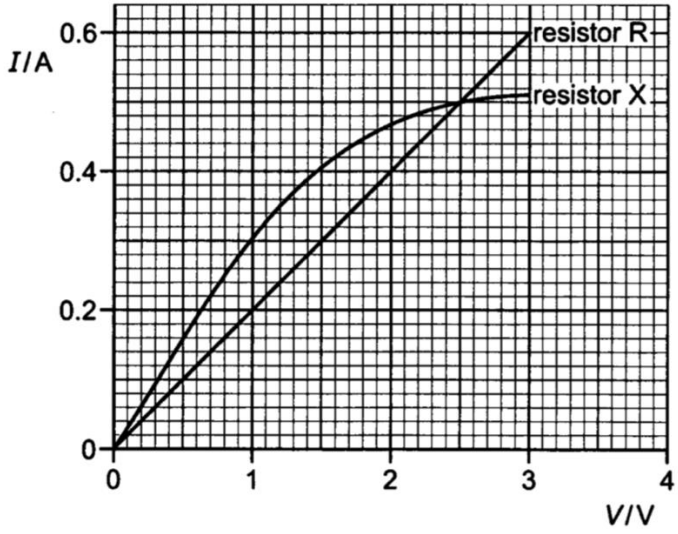

Fig. 4.1

(i) Student 1 sets up the circuit with resistors R and X in series across an unknown e.m.f. source with negligible internal resistance as shown in Fig. 4.2.

Fig. 4.2

Determine the value of the unknown e.m.f. source if the current through resistor X is 0.22 A.

Value of unknown e.m.f. source = …….. V [2]

(ii) Student 2 sets up the circuit using a variable resistor, resistors R and X as shown in Fig. 4.3 using the same e.m.f. source.

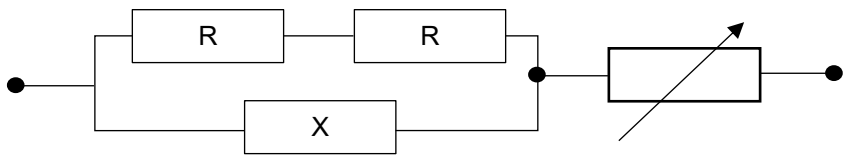

Fig. 4.3

Determine the resistance of the variable resistor given that the ratio of

$$\frac{\text{resistance of X}}{\text{resistance of R}} = 0.80$$

Resistance = …….Ω [3]

(iii) Student 2 modifies his circuit by adding an ideal ammeter and a voltmeter as shown in Fig. 4.4.

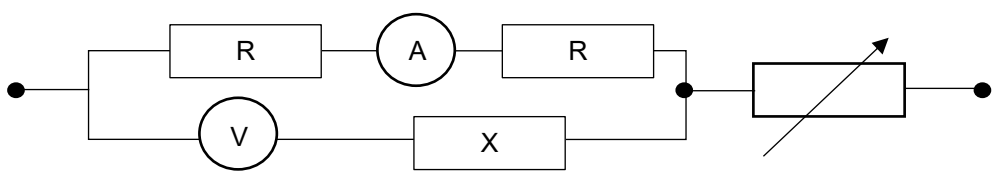

Fig. 4.4

[Turn over

Explain if this circuit is appropriate for determining the graphs over the range of V and I as shown in Fig. 4.1.

……

……

……

……

……[3]

4(b) A student uses the equation $I = Anvq$ to determine the drift velocity of the electrons, v, in the wire.
State, with a reason, whether this is the maximum velocity of the free electrons in the wire.

……

……

……[2]

5 **(a)** This question is about an americium-241 radionuclide source used in a